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REVIEW

Route planning methods in indoor navigation tools for vision impaired persons: a systematic review

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ABSTRACT

Objective: Route planning is key support, which can be provided by navigation tools for the blind and visually impaired (BVI) persons. No comprehensive analysis has been reported in the literature on this topic. The main objective of this study is to examine route planning approaches used by indoor navigation tools for BVI persons with respect to the route determination criteria, how user context is integrated, algorithms adopted, and the representation of the environment for route planning.
Methods: A systematic review was conducted covering the period 1982–July 2018 and thirteen databases.
Results: From the 1709 articles that resulted from the initial search, 131 were selected for the study. Route length was the sole factor used to determine the best route in the majority of the studies. Routes with less obstacles, less turns, more landmarks and close to walls were selected by the other studies. User variations were considered in few studies. Unique needs of the BVI persons were addressed by few novel algorithms which had integrated multiple parameters and BVI-friendly route modelling.
Conclusion: Differences between the navigation capabilities of the sighted and the BVI community were not a major concern when deciding the optimum routes in the majority of BVI indoor navigation tools. How to trade-off between factors affecting optimum routes, and how to model suitable routes in complex buildings needs to be studied further, looking from the user perspective.

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KEYWORDS
Visual impairment; indoor navigation; route planning; macro-navigation; algorithm

➤ IMPLICATIONS FOR REHABILITATION

- Navigation differences between sighted and blind and vision-impaired (BVI) communities are not concerned frequently when planning routes in indoor navigation tools of BVI persons.
- Selecting routes avoiding areas difficult to traverse, close to walls, having landmarks, and less turns are some approaches used to address the unique needs of the BVI community.
- Assisting for recovery from veering and real-time obstacle detouring are useful features offered by these tools.
- Identifying and prioritizing different factors contributing to better routes, concerning user variations, and adopting multi-objective route optimization will help to develop improved route planning methodologies for BVI indoor navigation tool

Introduction

With the limited capabilities to sense the world around, independent way-finding is a major limitation faced by blind and visually impaired (BVI) people. Way-finding involves two tasks; (1) sensing the immediate environment to avoid obstacles, and (2) navigating to a destination, which is far from the immediate surrounding [1,2]. Avoiding obstacles and moving safely in the immediate environment is a micro-navigation task; Navigating to a distant place, not in the immediate vicinity is a macro-navigation task. Macro navigation involves activities such as planning the route to take, and then following the planned route using micro navigation skills while updating the orientation.

Orientation and mobility (O & M) abilities, which strongly contribute to both micro and macro navigation activities are affected by impaired vision. Orientation refers to establishing a sense of where you are in the physical environment and mobility refers to how to move around safely. Therefore, orientation and mobility skills and guidelines specific to the BVI community have been developed over time to support BVI navigation, which has become the main strength supporting way finding and navigation for the BVI community [3,4]. Shore-lining, wall-tracing, avoiding large spaces, white cane techniques to detect obstacles at different levels of the surrounding, and sensing and familiarize with the environment to support developing mental maps are some of the O & M techniques [5].

The white cane is the primary tool used by the BVI community to safely negotiate the immediate environment and improve the O & M abilities. However, traditional devices have limitations such as the limited sensing range of the standard white cane [6]. Various assistive tools such as smart white canes, and orientation

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supporting aids, have been developed using different technologies to support the BVI to safely navigate with confidence [7]. Indoor navigation aids for the BVI, a branch of general indoor navigation tools, specifically support BVI people to navigate inside buildings. Any indoor navigation tool with full functionality will have common modules to handle positioning, route calculation and communication with the user [8,9], following the way-finding processes identified by Downs and Stea [10]. These modules are interrelated and depend on each other to provide navigation support. When calculating the route and developing meaningful instructions, building information is vital and managing that is also considered a main module [8]. The route calculation module of a BVI navigation system manages tasks such as planning optimum routes and constructing suitable navigation instructions based on the planned route. The route calculation module output is used by the communication module to convey the information in the most appropriate medium. The positioning module provides information to continuously track the BVI person's navigation in the environment so that information can be provided according to the user movements. Additional modules supporting orientation and obstacle avoidance strongly support BVI users and must therefore be integrated with route planning modules.

Many research and development activities have been done in the BVI indoor navigation tool domain; however, the actual use of such tools by the BVI community is still not prominent as highlighted by Gallagher et al. [9]. Due consideration for specific needs of BVI people is very important for the successful implementation of a navigation aid [3]. With regard to BVI indoor navigation tools, the different O & M abilities of the BVI community needs to be reflected in the overall system and the sub modules' design.

Several concerns specifically arise relating to BVI route planning; there can be indoor areas, which are difficult to transit such as cluttered corridors, which may affect the navigation of a BVI person; there can be unforeseen obstacles or environmental changes, and the user needs to deviate from the original route to avoid collision with them; veering is a common issue in BVI navigation, therefore support to recovery from veering can avoid getting lost; people with different levels of vision impairment may have different preferences for the optimum route to follow; environment features of the area the person is travelling is useful for developing a mental map therefore useful to be integrated to the navigation instructions. As the information requirements (e.g., knowing about obstacles and landmarks) and suitable routes (e.g., safe paths) are different for BVI people, the building representation also needs to be adjusted. Therefore, route planning becomes an important aspect of BVI indoor navigation tools, especially for complex or unfamiliar environments.

A survey on a special category of BVI indoor navigation tools, obstacle avoidance systems is presented by Dakopoulus and Bourbakis [11]. Gallgher et al. [9] evaluated several BVI indoor navigation systems using four criteria; device accuracy, device robustness, ability to integrate with a varying environment and information provided to the user. A short review of methods used in BVI navigation systems, mainly on the positioning is presented by Silva and Wimalaratne [12]. A review by the NRC in 1986 provides a classification of available systems and possible research directions based on user requirements, which are still important, especially regarding user expectations [13]. Surveys on general indoor navigational tools can be found in [2,8].

The main objective is of this study is to examine route planning approaches used by the indoor navigation tools for BVI persons with respect to the route determining criteria, how user context is integrated, algorithms adopted, and the environment representation for route planning.

Methods

The route planning support provided by indoor navigation tools for BVI persons are analyzed using a systematic review approach in this research. A systematic literature review is a process of identifying, evaluating and interpreting all relevant studies related to a particular research question and the need for such a review arises from the requirement of researchers to summarise all existing information about some phenomenon in a thorough and unbiased manner [14].

The standard guidelines for systematic reviews related to the health domain [15] cannot be adopted directly in this study as the domain is different; developing BVI indoor navigation systems is a transdisciplinary task related to orientation and mobility, assistive technology, engineering and computing. Therefore, the review protocol is designed following the guidelines from the software engineering domain [14,16] and the key points from the protocol are presented below. Amendments to the protocol during the review were agreed upon by all the authors.

Review questions

Review question formation

The following elements suggested by Kitchenham et al. [14] are used to derive the research questions: population - indoor navigation tools for the BVI community; intervention - route planning methods; comparison - no comparison; outcome – strategies related to route planning in terms of factors used in route planning, algorithms, user context, and the building data representation for route planning; and context - navigation of BVI adults inside buildings.

The review questions formed are: In indoor navigation tools for BVI persons, RQ 1– What route optimizing factors are adopted in route planning?, RQ 2 – What procedures and algorithmic approaches are used for route planning?, RQ 3 – How the variations in user needs are integrated to route planning?, and RQ 4 – How the building information is modelled to support route planning?

Review methods

Data sources

The articles related to the topic come from different disciplines including engineering, computing and disability studies. Following the Cochrane guidelines [17], to cover a wider search, sources are selected from bibliographic databases including subject-specific databases and citation indexes. The following thirteen sources are used for searching:

- Computing/Engineering: ACM Digital Library, IEEExplore Digital Library, Elsevier Ei Comp index, ProQuest - Electronics & Communications Abstracts/Computer and Information Systems Abstracts;
- Occupational health and social work/Health/Psychology: MEDLINE (OVID), EMBASE (OVID), CIRRIE, CINAHL, PsycINFO (OVID); and
- Multidisciplinary: Thomson Reuters - web of science Core Collection, SCOPUS, Springer, Science Direct.

Table 1. Keywords and synonyms used in the search strategy formulation.

Keyword	Synonyms
Indoor navigation	Indoor wayfinding
Vision impaired	Blind, blindness, diminished vision, impaired vision, low vision, visually impaired, visual impairment

Search terms and strategies

The keywords for a comprehensive search were identified based on a preliminary search in MEDLINE (OVID) where suggested related terms can be explored. The identified keywords and synonyms are listed in Table 1.

The search strategy is formulated as follows:

The key search phrase: (indoor navigation systems for vision-impaired people) – A

Vision-related key phrases: (vision impaired) OR (blind) OR (visually impaired) OR (low vision) OR (blindness) OR (visual impairment) OR (diminished vision) – B

Indoor navigation-related key phrases: (indoor navigation) OR (indoor wayfinding) – C

Combined logical search: (A or B) and (C). Although "route planning" is the main theme to investigate, it is only a sub-theme of BVI indoor navigation tools; hence, not used as a search term. The general search string used is: ("indoor navigation" OR "indoor wayfinding") AND ("vision impaired" OR "blind" OR "blindness" OR "diminished vision" OR "impaired vision" OR "low vision" OR "vision impairment"). Implementation of the search string was adjusted as per the requirements of different data sources.

Search scope

Time: 1980 is selected by Horton et al. [18] as the starting year in their review of the use of tactile technology in tools for the BVI community. Based on that, a preliminary check is done on ACM and IEEE databases and the first article satisfying search string is from 1982. Accordingly, starting year is set to 1982. The search period is from 1982 to July 2018.

Language: only the articles in English are considered.

Study selection

Inclusion criteria: Original research articles which present, (a) a design and/or development of a full or a sub-component, (b) a concept, (c) an evaluation or a requirement gathering, (d) spatial models, or (e) any other aspect, of indoor navigation systems for BVI community which concerns about the route to travel/route planning, are included in the review.

Exclusion criteria: Articles are excluded if they present content which met any of the following criteria: (a) research articles either published as survey or review or thesis, (b) original research articles which publish about navigation systems or concept related to navigation systems, (i) not linking with BVI persons, (ii) serving BVI persons but not for indoor, (ii) including route planning methods but not related to BVI indoor navigation, or (iv) supporting BVI indoor navigation but not consider about routes/route planning.

Accordingly, articles do not indicate route planning, such as articles only on localization methods or object identification, even though they are related to BVI navigation systems are excluded. While doing the screening, editorials and any unrelated content, articles on training systems or systems for children are added to the exclusion criteria.

Quality assessment

A quality assessment checklist is prepared to adopting the qualitative checklist presented by Kmet et al. [19] to the assistive technology domain, following the guidelines suggested by Kitchenham et al. [14,p.33]. Eight parameters relate to objectives, connection to context, method, design, evidence of implementation, the credibility of the evaluation process, conclusion and reflectivity of the study are used in the checklist. Items are given scores yes (2)/no (0)/partial (1) and the articles with a score of seven or more are included. The first and second authors performed the quality checking in a random sample to verify the process following the guidelines of Kitchenham et al. [14,p.33] and the first author conducted the quality checking of the rest of the articles. The first author is a PhD student in engineering and the second author is a senior researcher in computing. The quality assessment checklist and the score sheet can be found in the supplementary materials.

Study selection and data extraction process

Stage 1 (identification and screening): Each database is searched using the search string and the titles, publication year and the abstracts are extracted and any duplicates are removed. Results are screened based on the inclusion/exclusion criteria and relevant papers are selected. When the exclusion is doubtful, such articles are included for the next stage.

Stage 2 (checking the eligibility): Each full paper is accessed and checked for quality. Doubtful inclusions from the first stage are resolved. The resulting full papers are checked against inclusion/exclusion criteria and reasons for exclusions are recorded. The resulting papers become eligible records to be included in the review.

Stage 3 (data extraction): Following data items are extracted: title, year, paper type (proposal only/full paper), key features, route planning (Y/N), route optimization factors, route planning methods/algorithm, re-routeing support, user context and building representation information. Any other important factors are also noted. Though Kitchenham et al. [14,p.33] have recommended excluding multiple papers from the same data, papers from the same project are not excluded in this work as different articles can present different aspects or improvements.

The planning of the review, searching and screening of the data is performed using Parsifal (https://parsif.al/) software. Screening, article selection and data extraction is done by the first author. The second author performed article selection and data extraction in a random sample to verify the process following the guidelines of [14,p.33].

Data analysis and synthesis

Extracted data is used to answer the research questions using qualitative synthesis methods. The results are summarized under important headings in tabular formats and analyzed to infer findings related to each research question. Data analysis and writing of the manuscript were done by the first author. Both second and the third author reviewed the manuscript.

Included and excluded studies

A total of 1769 records was identified through the initial database search and 427 of them were eligible for full paper selection. From the eligible articles, 131 were selected for the analysis. A flowchart of the study selection process developed based on [15] is shown in the Figure 1.

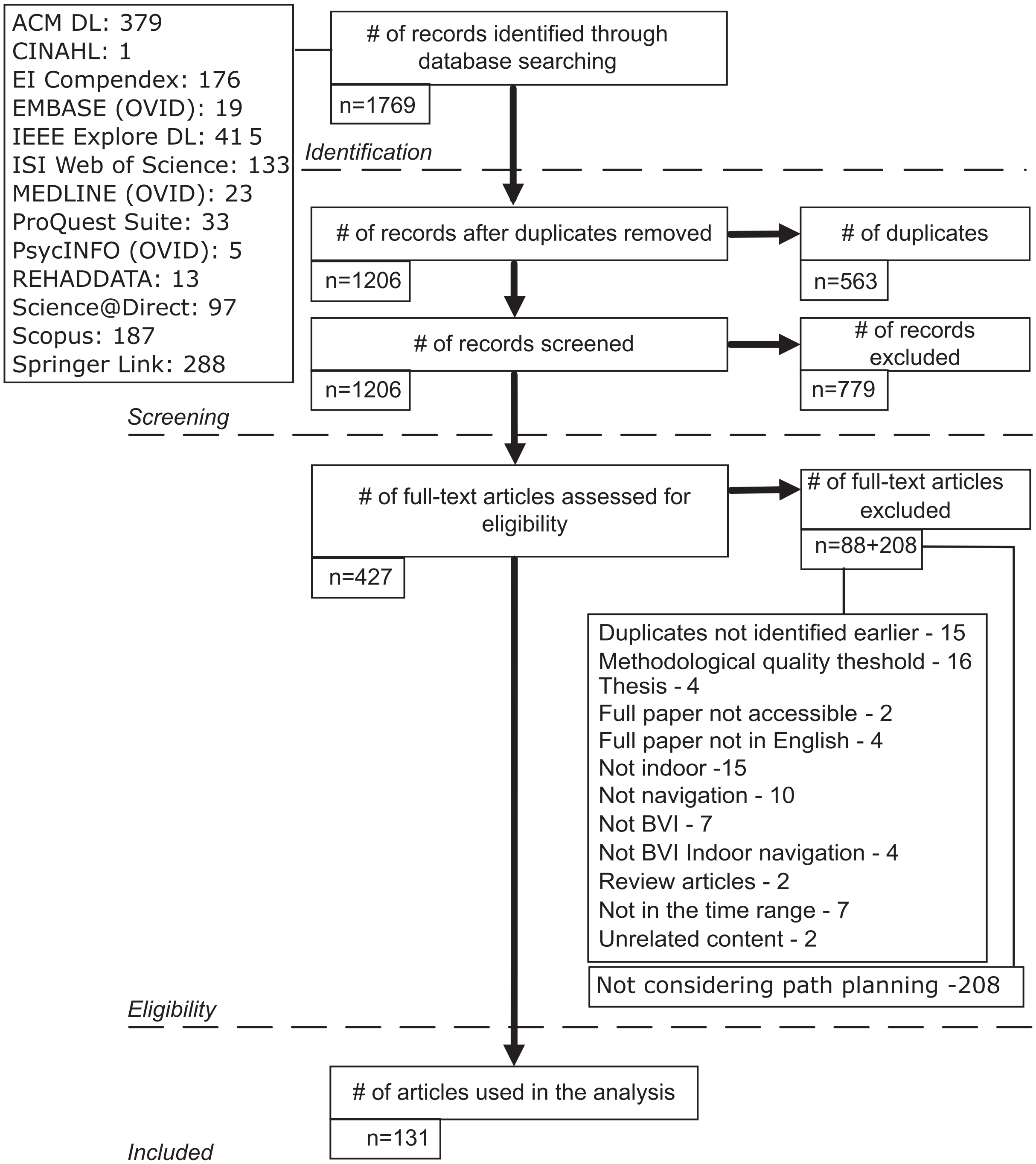

Figure 1. Flowchart of article selection process with exclusions for the review of route planning methods in indoor navigation tools for vision impaired persons.

Results

Overview

Out of 427 full papers resulting from stage 2, 131 articles were eligible for the analysis. Among the selected articles, 113 use a route planning method while 18 use pre-defined test routes. Two broad categories can be identified in the eligible articles (a) articles exploring specifically the route planning aspect [20–27], and (b) articles considering a navigation tool with macro-navigation support and including route planning as a sub-component (all articles except [20–27]). Navigation instruction formation with route modelling is focussed in two articles [28,29]. One article, [22], solved the itinerary optimization problem and all the others consider routes between two places. Guiding the BVI traveller with remote assistance is proposed by Scheggi et al. [30]. One article reports a user evaluation of an existing navigation system including the route planning aspect [31]. Selected articles are published from the year 2001 to 2018 and there is the gradual increase of the articles with time as shown in Figure 2. A considerable increase in articles considering the route planning aspect is noted after year 2014.

Results: RQ1 – route optimizing factors considered in route planning

Table 2 summarizes the different factors used as the basis for route optimization in route planning modules of selected articles

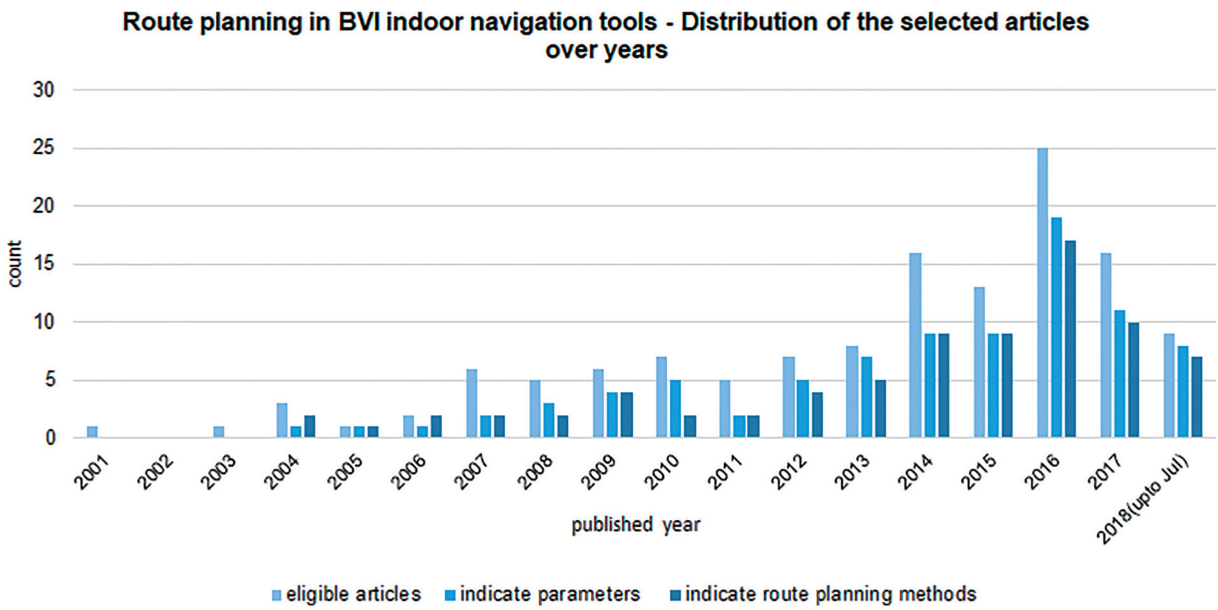

Figure 2. Distribution of the selected articles over years for the review of route planning methods in indoor navigation tools for vision impaired persons.

Table 2. Summary of factors considered in route determination.

Factor	Article reference
Distance	[22,23,31–69] (only distance) [24,70–79] (distance with real-time obstacle avoidance) [20,21,25,26,80–98] (combine distance with one or more other parameters)
Obstacles or difficulty in traversing	[20,21,24,26,70–76,78–80,82–85,88,89,92,93,99–102], [27] (arduousness of an edge), [77] (paths away from obstacles), [81] (obstacle, near obstacle, stairs/escalators, and near stairs/escalators), [86] (escalators), [87] (e.g., stair cases), [96] (e.g., obstacles on sides, no access to stairs)
Safe path	[71,103,104]
Landmarks	[21,26,84,85]
Turns	[26,81,83–85,91,94]
Rotation angle	[98]
Shore-lining support	[21,26,84,85,95,97,105] (priority for paths close to walls)
Crowd	[80,101,103] (close to the right wall to follow the flow of traffic)
Route complexity	[90]
Cognitive load related parameters	Number of doors to pass [26,84,85] number of intersection points in corridors [26,85] cognitive load to traverse [28] difficulties in finding the next landmark at the current position [95]
Environment details (e.g., lighting)	[25]
Sufficient space to walk	[24]

with article references. Sufficient details about factors used are not indicated in 44 articles (34%) therefore information from 87 (66%) articles are used in this section.

Travel distance

The most common approach for planning the route is calculating the route with the least distance to travel (see Table 2, row 1). 60% of the studies which mention a basis for route selection (52/87) use the shortest distance as the sole criteria while 11 of them adjust the route with real-time obstacle detouring. The shortest distance is considered coupling with other parameters in 23 articles.

Obstacles and turns

Routes, which have less inconvenient features such as turns [26,81,83–85,91,94], hazards [99], obstacles on the route [20,21,24,26,70,71,74,80–85], and the magnitude of the rotation the BVI user has to make when coupled with a robot [98] are selected in some studies. Two approaches are used to incorporate these factors in route calculation; (1) assigning a different weight/penalty for the route segments with obstacles [26,80,81,83–85,92,93] or turns [26,81,83–85,91] and, (2) identifying the areas with obstacles and avoiding those areas completely [20,21,82,88,89]. The "degree of difficulty to traverse each link", which is a combination of distance and the obstacle presence is used by Ganz et al. [80]; if there are obstacles (e.g., furniture) along with the link, a higher weight is assigned while an infinite weight is assigned if places of danger (such as very crowded area) are along with the link. Though the distance is used for finding the route by Lee and Medioni [77], the route is adjusted to sufficiently away from obstacles.

Real-time obstacle detouring is addressed in several studies [24,70–79,100–102]. It is common that obstacles in 3D space are reduced to 2D information in route determination. Two relatively recent (published in 2014, 2015) articles consider obstacles in the 3D space [24,70]. While avoiding the obstacles, sufficient space to traverse by a human is also checked by Díaz-Vilariño et al. [24]. Some studies not only avoid the obstructed route but increase the safety by setting some distance from obstacles as well [71,82]. Tables, desks and bookshelf [83], walls, pillars and furniture [71], chairs, tables and crowded area [80], furniture and columns [24], staircases [87] are objects noted specifically as obstacles.

Safest route

Safest route is the main concern of the proposed route planning modules by Chippendale et al. [103] and Árvai [104]. Acknowledging that different factors can increase the safety, routes densely covered by obstacles or with limited accessibility (for guide dog users) such as revolving doors and escalators are planned to be avoided in [104]. A less crowded route is considered safer than a crowded route in [103]. Mekhalfi et al. [71] have assumed that a safe route is a route without obstacles. Escalators, which can be a safety concern, are avoided by default in route selection by Sato et al. [86].

Landmarks

Landmarks alone the route are considered a deciding factor in selecting a route in four studies (see Table 2, row 4). The number of landmarks along a route is one of several parameters used in calculating the optimum route in multi-parameter navigation model presented by Ivanov [84], Ivanov [26], Ivanov [85] and Swobodzinski and Raubal [21].

Supporting applying O & M techniques

It is notable that very few studies consider support to applying orientation and mobility skills such as shore-lining in route determination [21,26,28,84,85,95,105]. The difficulty in finding the next landmark at the current position is incorporated into the route planning in [95] and a higher cost is given for the edges when crossing a hallway than when following an adjacent wall, as it is more difficult to cross a hallway than following a wall. A user centric-navigation graph to represent the route to follow is generated by Dong and Ganz [28]. The actions taken by BVI users when navigating are assigned to each link of the graph. The cognitive load to travel each link is used to determine the suitable route in [28]. Oh et al. [95] have suggested that the paths suitable for the BVI community be generated by authorized users such as building managers and orientation instructors so that the routes more supportive to BVI users can be identified.

Routes following walls are specifically selected first to derive the optimum navigation route in several studies [21,26,28,84,85,95,105]. Though a route, which follows not the middle of the corridor but the edges of the hallway is specifically selected to support the shore-lining techniques the route is manually pre-determined for the route following evaluation by Richle et al. [105]. Hosney et al. [81], have noted that the paths near walls would be preferred by some users while others may not like them, though this idea is not used in route planning.

Considering the fact that tactile markers can be used as a guided route for white cane users, existing tactile routes on the floor [86,106,107] and newly introduced tactile guidelines [49] are used as routes. Coloured lines conventionally marked on the floor of public places are also used as routes [108].

Multi-objective route planning

More than one parameter used to obtain better routes for BVI people can be identified in 33 articles as shown in Table 3. Distance is coupled with another parameter such as the number of obstacles or the number of turns in most of them.

The prioritizing of the multiple parameters is commonly done as an *ad-hoc* task. Hosney et al. [81] have proposed to add a penalty factor for three “undesirable features”; obstacles, stairs/escalators and turns in the route, for BVI wheelchair users. Different weights are given for direct links, junctions, and indirect links by Simões and de Lucena [91].

Integrating several BVI requirements, Ivanov [84] has used five parameters, namely (1) number of landmarks on the route, (2) number of obstacles and hazards on the route, (3) number of doors through which the user must pass, (4) number of turns, and (5) the length of the route, in deciding the optimum route, which is selected specifically from the set of routes following the walls. However, what weights are assigned to each parameter is not reported. Extending the same study, another parameter, the number of intersection points in corridors, is added later by Ivanov [26] and Ivanov [85]. Selection of the parameters and scoring of the criteria is done based on rank correlation coefficients calculated using data from an interview with 25 BVI people and 5 experts in [26]. Each individual has given the rank for each criterion as per their preference. The weights for each parameter, length (m), number of obstacles, number of landmarks, number of doors, number of turns, number of intersections are 0.2640, 0.2360, 0.2267, 0.1507, 0.0853, and 0.0373 respectively as reported by Ivanov [85].

Arguing that a route, which supports (1) wall-trailing, (2) avoiding obstacles, and (3) having landmarks and cues along with it, would result in a safer travelling than a route with the minimum distance, these three parameters are coupled with distance by Swobodzinski and Raubal [21]. The landmarks are assigned twice the weight of cues (based on the count) in this model and the route with the higher weight is selected if its length is less than twice the length of the route with the lesser weight. Swobodzinski and Raubal [21] have emphasized that, as there is no available information related to prioritizing the distance over other factors, this simple method is adopted.

Routes which maximize usability and minimizing the chances of getting lost are suggested to be calculated by Kahale et al. [27], where the edges are assigned weights based on two parameters; physical constraints of the user, and the intrinsic features of the edge. Sato et al. [86] have avoided escalators and include routes with tactile paving for users with visual impairments by default. The cognitive load to traverse each link is used as the weight and the actions are assigned to links, selecting from nine primary actions and 13 secondary actions by Dong and Ganz [28]; however, what weights are assigned is not presented. Context-aware navigation models which use several contexts including the environmental context in route planning are proposed by Fernando et al. [25] and Wu et al. [109].

Though multiple parameters are discussed, how the planning is done based on them is not reported in several studies [25,90,96,99,101,109]. Idrees et al. [94] have provided two choices, routes with the least distance or routes with the least turns, but did not integrate the two parameters.

Results: RQ2 – addressing user variations in route planning

In the majority of articles, a general BVI user is considered in route planning and no adjustments are made based on individual user preferences or variations in capabilities. Differences in users are considered in a limited number of studies (15 out of 131) and they use two approaches for integrating the user profile to route planning; (1) allowing the individual user to provide preferences, and/or (2) defining rules based on user groups. The different attributes used to define user-profiles and preferences are listed in Table 4.

Users are allowed to provide preferences by selecting the parameters or adjust the weights for the parameters in the route selection [81,86,90,92,94,109,110]. Selecting either the shortest path or the path with least turns [94], preference of objects to

Table 3. Combination of multiple parameters adopted for path planning.

Article reference	Distance	Obstacles/ hazards	Turns/rotation angle	Landmarks	Doors to pass	Intersections	Close to walls	Route complexity	Other
[20]	✓	✓							
[21]	✓	✓		✓			✓		
[24]	✓	✓#							
[25]	✓			✓					✓ (Environment)
[26]	✓	✓	✓	✓	✓	✓			
[70]	✓	✓#							
[71]	✓	✓#							
[72]	✓	✓#							
[73]	✓	✓#							
[74]	✓	✓#							
[75]	✓	✓#							
[76]	✓	✓#							
[77]	✓	✓#							
[78]	✓	✓#							
[79]	✓	✓#							
[80]	✓	✓							
[81]	✓	✓	✓						
[82]	✓	✓							
[83]	✓	✓	✓						
[84]	✓	✓	✓	✓	✓				
[85]	✓	✓	✓	✓	✓	✓			
[86]		✓							✓ (Tactile paving)
[87]	✓	✓							✓ (Time)
[88]	✓	✓							
[89]	✓	✓							
[90]	✓							✓	
[91]	✓		✓						
[92]	✓	✓							
[93]	✓	✓							
[94]	✓		✓						
[95]	✓						✓		✓ (Difficulty in finding the next landmark)
[97]	✓							✓	
[98]	✓		✓						
[99]	✓	✓	✓						

= Real time obstacle detouring.

use as desired destinations [110], mobility preference [86], willingness to use staircase [90], and floor crossing preference (staircase, elevator or escalator) [92] are some factors allowed to be set by the users in route planning. Wu et al. [109] have concerned about integrating user preferences in route selection as a key requirement. Allowing the users to change the weights of the cost function parameters is suggested as enhancements by Hosney et al. [81] and Ivanov [84]. Though the individual preferences are not considered, the need for such approach is highlighted based on a user evaluation by Ganz et al. [34]. Users are allowed to set the preferred distance from the wall by Muñoz [75]. Kulukin et al. [111] have proposed to allow user preferences in the route selection however no details are provided.

When defining group-based rules, vision or other mobility-related characteristics are used to categorize the users [25,27,81,84,85,87,90,112]. User disability is used as a parameter in selecting the optimum path by Rodriguez-Sanchez and Martinez-Romo [96], where different forms of disabilities including vision impairment are considered. Three categories of disability-related user characteristics affecting navigation are identified by Kahale et al. [27]; disabilities related to use of hands, displacement and vision (five sub-variations). Combining variations of vision capabilities with physical limitations is proposed by *Fernando* et al. [25]. Three general, two location-related and six user characteristic-related parameters are used by Ivanov [85], and the impairment type is one user characteristic identified. Vision impairment is identified as an impairment type. It is notable that the individual visit history and deviations are to be recorded in [85] so that they can be later used for better support. Context-aware navigational models [25,84,109] and BVI user navigational ontologies [90,112] are some approaches used to integrate differences in user perspective.

Avoid the stairs if a wheelchair user [90], avoid the escalator if accompanied by a guide dog [87], use the escalator instead of the staircase if only a white cane is used [87], and move away from walls if a BVI wheelchair user [81] are some rules adopted to integrate user capabilities. The combination of the personal profile of the user and a user group convenience rate is proposed to be used when selecting a route by Völkel and Weber [87]. They have suggested that a higher convenience rate can be used if a path is more often used by different users. Though a comprehensive user profile is maintained by Ivanov [84], how it is used in route planning is not indicated.

Results: RQ3 – procedures and algorithms used for route planning

Algorithmic approaches used in route determination in the selected articles are summarized in Table 5. Information about the route planning methods is presented in 79 studies out of 131.

The most common algorithm adopted for finding the suitable route is Dijkstra's shortest path algorithm [121] and the A*algorithm [122]. Heuristics such as turns and Manhattan distance are used with the A* algorithm. Dijkstra's shortest path algorithm is the choice of 33 articles (42%). D* and D* Lite algorithms are used to support dynamic route planning. The articles which focus on route planning methods present improved versions of existing algorithms, adjust the weights of the criteria

Table 4. User characteristics considered in route determination.

User characteristics considered	Details and the article reference	
Vision-related parameters	Vision [109]	
	related to vision [27]	
	◦ blindness (use of a walking cane)	
	◦ glare, fog, blur, opacity, paleness	
	◦ fragmented, tunnel or peripheral vision	
	◦ blindness (use of guide dog)	
	◦ difficulty reading	
	impairment-type (none, blind, visually impaired, mobility) [85]	
Mobility difficulties and use of supporting tools	Guiding tools, walking pattern/habits (speed, stride) [109]	
	related to the displacement [27]	
	◦ supporting stick, crutches, walker	
	◦ stiffness, joint or muscle pains	
	◦ shortness of breath, respiratory or heart problems	
	◦ electric wheelchair	
	◦ manual wheelchair	
	impairment-type [none, blind, visually impaired, mobility] [85]	
	impairment-type [none, blind, mobility] [84]	
Personal information	Age [109], interest [109]	
	name, gender, type (visitor, staff) [84], [85]	
	social-networks (Facebook, Twitter) [84]	
Navigation route preferences	Preferred-spatial-representation (allocentric, egocentric)	[85]
	preferred-path (min-path-length, min-number-of-turn-points, max-number-of-landmarks, min-number-of-obstacles, min-number-of-doors)	
	Preferences [spatial-representation, route- weights]	[84]
	Location (user location (path element id))	[84,85]
	Position (3D user position)	
	activity-status (stay, walk, run)	
	Deviation-history (normal, often, too-often)	[85]
	History (date, building-id, route-elements-id)	[84]

Table 5. Algorithms and approaches used in BVI route determination.

Algorithm/approach	Article reference
Dijkstra's shortest path	[28,33,36–39,41–46,51,55,67,71,74,80,83,92,95–97,101,111],
	[60] (Modified version) [31,93] (path planning is as per a previous paper)
	[35,47–49,59] (Shortest path algorithm)
A*	[81] (Heuristic: Manhattan distance)
	[94] (Heuristic: Turns or Distance)
	[82] (Heuristic: Manhattan distance)
	[75] (Heuristic: L2 image distance, results smoothed)
	[54] (Heuristic: Manhattan distance; Douglas–Peucker algorithm for reducing points)
	[52,56,65,68,88,98]
BFS	[32,101,113–115] (all these are related to a continuation of a same project on a robotic shopping assistant), [50,58]
D*	[79] (CAMShift for obstacle detouring)
Dijkstra and the Floyd–Warshal algorithm	[61]
A* and D* Lite	[53] (hierarchical path planning)
A* and jump point search	[62] (VR system)
Single path calculation method	[89] (addressing low memory issues of the hardware)
Ontology and rule-based approach	[90]
Optimisation methods	[23] Particle swarm optimisation
SLAM-based approaches	[66,116] (Dijkstra for the global planner)
	[64,76,117] (A* as the global planner),
	[70,77] (D* Lite as the global planner, with adjustments)
	[100] (A* as the global planner with post processing)
	[78,118]
Novel approaches considering BVI needs	[20,21,24,26,27,84,85]
	[22] (solving TSP problem using the Christofides algorithm, nearest neighbour algorithm and the Kruskal algorithm)
Other	Bayesian algorithms-partially observable Markov decision process [119]
	Room to room paths based on access to rooms and room centroid [40,120]
	Remote-guidance [30]
	Visual semantic parameterisation [72]

based on BVI needs or develop novel route planning approaches considering the unique needs of the BVI community (see Table 5).

Solving itinerary optimization in a supermarket, Khalifa et al. [22] have reported that using Christofides algorithm [123] and nearest neighbour algorithm, addressed using Kruskal algorithm [124] provides a feasible but not optimum solution for finding shopping items. As this approach can include already visited shelves again, it is improved to go through all items existing in the same shelf and to partitioning the shelf areas to regions and thereby use item category and reported achieving successful results.

Route planning is done as an iterative process by Serrão et al. [40]. Which access structure is neighbouring to the starting region, in the same direction, and the shortest distance to the destination is first checked. Then from that access structure, the next room it connects is found and the room becomes the new starting region. The process is iterated until the destination is reached. If a region is found without further access structures to connect to other places, the proceeding structures are marked as invalid for the present route, the algorithm returns to the initial state and proceeds with the new neighbourhood maps. Using this process a route, which connects the centroid of the rooms *via* doors/staircases is found. However, whether an access structure such as a staircase is preferred by a BVI user or not is not considered in the route planning. The same idea is used in an extension of the project by Serrão et al. [120].

Yosuf et al. [23] have used particle swarm optimization [125] in a simple scenario, presenting the only study which has adopted an evolutionary algorithm approach for BVI route planning. The dynamically calculating route is pruned and simplified with lesser nodes to generate more straight lines in the route by Muñoz [75].

Novel algorithms specific for BVI navigation

A unique methodology for calculating optimum routes considering orientation and mobility principles is adopted by Ivanov [26] (with related articles [84,85]) and by Swobodzinski and Raubal [21]. They assume routes along the walls are better than other routes; hence, the environment is modelled accordingly to create the search space of the navigation graph (see building data modelling section for details).

Swobodzinski and Raubal [21] have developed the route calculation based on the idea that if the route can consider the principles of white cane use, it would be better for a blind person in terms of safety and orientation than a route, which considers only distance minimization. Highlighting the significance of walls in BVI navigation for trailing, alignment, self-familiarization and orientation, routes closest to the walls are prioritized in [21]. The novel algorithm in [21] finds routes, (1) supporting wall-trailing, (2) avoiding obstacles, and (3) having landmarks and cues along the route. Distance is coupled with those three parameters and if the resulting optimum path's distance is more than twice as much as the shortest path, the optimum path would be ignored. It offers two versions; one for the sighted person and one for a blind person (see [21] for the pseudo-code). In the blind person's version, first, a buffer of 90 cm is derived along the walls to support a route facilitating full cane sweep and then, from the start and end points, straight lines are calculated towards the buffer if any of these points are not within the buffer. Then two routes, which can initiate either clockwise or counter-clock-wise direction are calculated. If obstacles are found along the route, a buffer is created again, and a new route corpus is generated. Finally, a utility score based on the number of landmarks or clues along the route and the distance is used to select the optimum path.

Ivanov [84] proposes to arrive at the optimum route in two steps; first a "coarse estimate", which indicates the sequence of rooms and doors which the user needs to pass, and then the "accurate estimate" which finds the optimal route within rooms and hallways are calculated. The second step manages the obstacle detouring and wall following by finding routes, which are 50 cm away from the walls and fixed obstacles (see the section on multi-objective route planning for details of the parameters used in [84]). The same idea is extended by Ivanov [26] with three stages: (a) finding the k-shortest paths to a given target, (b) finding fine-grained paths using a geometric model, and (c) selecting the highest rank path based on a multi-objective optimization method. The weight of each route segment calculated using six parameters is aggregated to rank the routes. In the second stage, in order to find paths adjoining walls and detouring obstacles, a novel buffering method is derived which can generate a polygon and polyline buffer around the walls and objects having both convex and concave shapes. The resulting paths are all along walls, detour obstacles and derived with enough space to traverse. The actual geometry of the room and the obstacles within the indoor space as well as the complexity of a large building are considered in this approach.

Obstacle detouring to support BVI navigation is considered as the key objective in the route planning methods proposed by Wu et al. [20], Díaz-Vilariño et al. [24], and Lee and Medioni [70]. A visibility graph approach is used in [20] with a newly derived data structure termed a "cactus tree" for the environment modelling. The route planning method in [20] consist of three parts; cell decomposition, the cactus tree-based planning for top-level plan and the Dijkstra/A* based route planning for small areas such as rooms. How to incorporate the dynamic changes of the surrounding to route recalculation is considered as the main concern in route planning in [70] and the D* Lite algorithm which supports such adjustments is selected for route planning. As the subtle changes in the map may cause frequent changes in the route, which would be confusing to the blind user, reliable waypoints, which are adjusted waypoints of the originally planned route, are used in [70].

Route planning is performed in 3D space rather than the 2D space by Díaz-Vilariño et al. [24]. First, a navigable network is generated as a variable density network based on a Voronoi diagram and this dual graph is used for route planning. When generating a route from a known location, it is added to the network and a new Voronoi cell is generated. The Dijkstra's algorithm is run on this navigable network to find the shortest path, while a 3D rectangular buffer is displaced following the path to check if any obstacles are found. The network is changed and new nodes are added if obstacles are found. This new network is used again to select the path, and the process continues until the destination is reached.

Hierarchical path planning

Hierarchical path planning helps to reduce the search space. Some studies considering BVI navigation in large or multi-story building areas adopt hierarchical approaches for route planning [20,53,83]. By differentiating between moving from one floor to the other, movements within a floor and within a small area of a single floor, only the required space is used for the searching in the systems presented in those articles.

Navigation planning is done in three stages by Martinez-Sala et al. [83]; planning entrance to the floor, within the floor to the desired bookshelf, and from the shelf's end to the desired item. Movement between the floors and within a floor is solved using two different approaches as no fine details are required when moving from one floor to another. Similarly, a high-level planner for building-wide and multiple floor connectivity (A* algorithm) is used while a low-level planner is adopted for room levels (D* Lite algorithm) by Kannan et al. [53]. The high-level planner provides a restricted route for the grid level planner so that the search space is restricted and the computation overhead is reduced. The grid planner is optimized based on two heuristics. They reported a significant drop in re-planning time compared to D* Lite algorithm alone. Two approaches for floor level and between regions are used by Wu et al. [20]. Floor level search is addressed based on a new data structure while the Dijkstra's/A* algorithm-based searching in a visibility graph is used within a region in [20].

Breaking down the navigation task into manageable steps, especially when navigating in complex environments such as multi-story buildings and floors with large areas, reduce the cognitive load of the users also. Placing kiosks at strategic locations such as the entrance of the building or near an elevator is the main approach used to reduce the cognitive load [34,48,95,126–129]. The routes from a kiosk to the next kiosk or the destination are planned instead of the complete route at once in these systems. Interactive tactile maps [127,128], a service-robot acting as an information desk clerk [129], a computer system, which transfers data to the user's phone [95,126], and special RFID tags-based interface [48] are used at kiosks.

Dynamic route calculation and re-routeing

Recovery from veering and re-routeing is supported in sixteen articles [44,46,47,55,73–75,79,82,84,86,90,101,102,110,130]. Different approaches are used by these studies to handle the veering problem. The most useful approach is the automatic re-routeing with veering detection as implemented by Cheraghi et al. [55], Muñoz et al. [75] and Sato et al. [86]. Legge et al. [130] let the user find a close-by tag if get lost and recalculate the route from that place. The user is allowed to change the destination and request rerouting anytime during the journey by Manlises et al. [79]. Instead of recalculating the route, Idrees et al. [94] guide the user to a previously known correct location if a deviation occurs. Kassim et al. [69] detects the veering based on a digital compass and continuously alert the user *via* a warning message, expecting a user to turn and take the right direction himself. The route is adjusted in real-time dynamically to detour obstacles in several studies [24,70–79,100–102].

It is common that obstacles in 3D space are reduced to 2D information in route determination. Two relatively recent articles (published in 2014, 2015) by Díaz-Vilariño et al. [24] and Lee et al. [70], have considered obstacles in the 3D space. Complementing the functionality of the white cane, obstacles in the 3D space, which would not be possible to detect *via* the white cane are considered in [70]. Using point cloud data, 3D traversability of the immediate environment is identified, a 3 D local voxel grid map is created and the route deviations are calculated to avoid collisions with dynamic obstacles in [70]. A rectangular prism-shaped buffer which represents a person is moved along the route and check for the obstacles in [24].

Computer vision-based methods with wearable sensors [72,75,76], robots [102], Radio Frequency Identification (RFID) based methods [73], multiple data from vision based sensors and IMUs [71] are used to manage the real-time obstacle detouring. Re-routeing the user dynamically to avoid crowded corridors/areas is proposed by Chippendale et al. [103]; however, the design or the implementation is not presented.

RQ4: building data modelling for route planning

Indoor route determination methods are strongly coupled with the way the built environment is represented. Therefore, the studies which look at route planning approaches to fulfill the specific needs of the BVI community concerns about building data representations. Both the cell-based [49,50,54,73,74,79,81–83,88,89,100,117], and the node-edge navigable graph-based [22,28,33,35,37–39,41,43–45,48,51,55,60,61,64,65,67,72,75,76,80,91,92,94,95,97,104,106,111] approaches are common in environment representations for route planning.

What positions are used as nodes are important to derive suitable, safe routes as well as navigation instructions. Rooms [48], passage points along the corridors to destinations [35], RFID tags locations [69,74] (placed on the floor), QR code locations [94] (placed on the floor) [67] (placed at room locations, etc.), stop blocks of braille block networks [106], landmark locations which are then marked with NFC tags [80,92], visual markers [51,91,97], points of interests (POIs) [64,65] or doors [63] become nodes of navigation graphs frequently. The routes following walls are first derived to generate the navigation network by Ivanov [26] and Swobodzinski and Raubal [21].

How the environment is tessellated is important to capture a sufficient level of information for route planning in grid-based approaches. Rectangular grids are commonly used with varying grid sizes such as $0.4 \times 0.4\,\text{m}^2$ [73,74,83], 1.5×1.5 m^2 [49], 0.5×0.5 m^2 [54], 1.44×1.44 m (area of four tiles) [88], 0.801×0.801 m^2 (area of four tiles) [82].

Kannan et al. [53] have varied the environment representation based on the level of details in their hierarchical route planning approach. Graph-based representations are used for high-level maps where connectivity between floors and rooms is maintained and grid-based representations are used for connectivity within rooms and hallways. The visibility graph approach is used to produce the navigation graph by Wu et al. [20] and Kahale et al. [27].

Voronoi polygons are used for environment tessellation by Mekhalfi et al. [71] and Díaz-Vilariño et al. [24] with a view to support obstacle avoidance. Initial points triggering Voronoi polygons are the nodes representing doors and concave corners in [24], while the varying level of densification is used to add additional points to different areas. Voronoi polygons are created with the boundaries of the fixed obstacles as starting seeds of cells in [71]. Then the start and the destination points are connected to the nearest cell based on the Euclidean distance and the safest route is calculated based on the Voronoi diagrams' edges. Legge et al. [130] have tessellated the floor space (base polygons) with triangles in such a way that no triangle edge crosses a wall.

Wu et al. [20] have proposed to use an "intelligent map", which is a model integrating the schematic and the semantic concepts of the indoor environment. It uses three map layers; topographic, schematic and semantic. Importantly, the distance between navigation nodes is kept less than five or six metres following the empirical observations and RNIB (https://www.rnib.org.uk/) guidelines, as mentioned by Wu et al. [20]. In order to realize this information for navigation, a novel data structure called a "cactus tree", which maintains pre-knowledge about relationships between indoor elements of the intelligent map, is proposed. It is a non-linear data structure represented as an ordered/sorted tree. Instead of simple before and after relationships, it maintains left, right and branches (below or equal) relationships as well.

3D building modelling for BVI navigation is considered by Díaz-Vilariño et al. [24], Fallah et al. [52] and Lee and Medioni [70]. A 3D virtual representation of a building is adopted in [52] and the model is converted to a 2D map for route planning. 3D local voxel grid map based on point cloud is used in [70] though the route planning is done in 2D space.

Limitations of the network models for BVI navigation are highlighted by Swobodzinski and Raubal [21] and Ivanov [84]. Arguing that the network models, where "room-edge" relationships are simply converted to "node-edge" relationships, are not suitable when dealing with a large number of nodes, Ivanov [84] has proposed a non-network based model. Ivanov [84] has suggested to use a semantically-rich object-oriented modelling of building data coupled with sensor data as these data are needed for the selection of a suitable route. The main limitations of a network-based models identified by Swobodzinski and Raubal [21] are, (1) restricting the movements to static pre-defined routes, (2) different possibilities for creating a network, (3) the lack of explicit obstacle avoidance, and (4) the problem of route optimization suitable for BVI navigation. To overcome the limitations, Swobodzinski and Raubal [21] have specifically defined the environment to support applying the orientation and mobility principles.

Use of standard building models

Standard building models or spatial data formats are adopted in a small number of studies [21,24,38,48,90,100,106,112]. Use of shape files and spatial databases [48,90], extracting data from available drawing files [38,100], use of IFC (https://technical.buildingsmart.org/standards/ifc/) as the building model [48,90,112], are some methods adopted in GIS-based representations. Ryu et al. [106] have extended the IndoorGML (https://www.ogc.org/standards/indoorgml) to cater for BVI navigation and the landmarks are added to the edges of the network, with four directional topologies (left and right side on the route and the left and right side over the route). To form navigation instructions based on the direction of travel, a directional graph is created by splitting the junction nodes of the network. 3D point cloud building data is stored in gbXML (https://www.gbxml.org/) format by Díaz-Vilariño et al. [24], justifying that it is sufficient for generating the navigable maps.

Semantic building data representation

Though the topological data is used for routeing, the semantic information can improve the routeing decisions and instructions. Semantic information is modelled in detail in few studies [26,42,84,85,131]. Cecílio et al. [42] have stored information such as floor number, relative positioning, and wall condition and modelled the data in different levels of details such as rooms at the top level and POIs within the room at a lower level. Bilgi et al. [131] have stored the information under six categories with descriptions; exits, emergency equipment & evacuation, restrooms, information desk, food courts, and ATMs. A semantically rich object-oriented data model is proposed by Ivanov [84]. The same idea is extended by Ivanov [85] with eight building-related classes (building, room, door, window, stairs, elevator, object and wall), their attributes and relationships to effectively use by a BVI indoor navigation model. Attributes detailing route width, gradient, steps, stairs, elevator, escalator, and tactile paving of each edge of the graph are used in the spatial graph created by Sato et al. [86].

Sample environments considered. General or specific scenarios: While most of the systems consider the indoor spaces in general, route planning for specific scenarios is addressed in several articles. Shopping malls [22,32,42,86,88,103,115,132], a university library [83] (up to selecting an item), an airport [107], a transit station [46,133,134], a museum [31,49,135], a hospital [127,128], and a building in a campus [131] are some of such environments. Instead of planning a route from one source to one destination, Khalifa et al. [22] have addressed the itinerary planning problem in a supermarket where different items are to be purchased in a single visit. An emergency response scenario is considered by Penmetcha et al. [50]. Both indoor and outdoor environments are considered by few studies [96,99,131,136].

Complexity of the sample spaces considered: Simple environments such as a room [89] or a section of a floor [39,51,82,91] or a single floor [41,76,94,95,97,100] is commonly selected as the sample area. Route planning including the transition between two floors are considered in few studies based on multiple floors of a building [28,34,38,48,52,53,86], two floors of a campus building [55,67], and two floors of a council building [96]. Multi-floor route planning with floor detection is specifically addressed by Muñoz et al. [75]. A large floor area with obstructions is considered as the sample environment by Au et al. [37], Mekhalfi et al. [71] and Oktem et al. [88]. Few studies have used two sample areas, one simple, and the other complex [96,101]. Highly complex environments are used for the deployment of the tools by Kim et al. [46] and Sato et al. [86]; a shopping mall with 21,000 m^2 area of three floors is used as the sample environment and for real time evaluation with BVI users in [86] where sample routes with 50–177 m lengths are used; the system developed in [46] is evaluated in a busy, crowded and a large railway station servicing more than 400,000 passengers per day, with routes of lengths 114–171 m. Though implemented in a multi-story shopping mall, details about navigating between floors are not presented by Athira et al. [137].

Discussion

This systematic review examined and reported the route planning methods used by the indoor navigations tools developed for BVI people, specifically looking at the route planning criteria and methods used, how the user context is integrated, and how buildings are represented for navigation.

Key findings: RQ1 – route optimizing factors considered in route planning

Distance is the only route optimization factor used in the majority (60%) of studies. Minimizing distance together with obstacle avoidance is common in many of the remaining articles. Safe paths, turns/angle to take, landmarks and cues, floor-transit features in the path shore-lining support, crowd, route complexity, cognitive load related parameters such as difficulties in finding the next landmark at the current position, sufficient space to walk, and environment details such as lighting level are the other factors used to identify optimum routes for BVI indoor navigation.

When the special concerns of the BVI community regarding route choices are not a priority of the system design (e.g., priority is for localization) least distance is a reasonable choice for route planning. However, when only distance minimisation is considered, other factors which contribute to the safe and efficient navigation of BVI persons are not accommodated in the resulting route. For example, some BVI users may prefer to avoid crossing large empty areas or crowded paths through the travel distance is increased.

Navigation differences in sighted and the BVI persons are strongly considered and support to applying O & M strategies of BVI persons such as wall-following, avoiding crossing large empty areas, and reducing number of turns are treated as important factors by few studies [21,24,26,80,81,83–85,90,91,94,95,103,105]. Overall, the impact of the differences in BVI and sighted people's capabilities on optimum route selection is poorly acknowledged, though the route planning is considered in 131 articles.

Landmarks are important features for BVI navigation which can serve many purposes; (1) support localization by verifying the user position against the landmarks, (2) use in navigation instructions (opposed to or complement the turn by turn navigation), and (3) support establishing orientation and mental map building by knowing the presence and relationship between landmarks. In order to use landmarks for any of those purposes, they should be close enough to the walking path to recognize them *via* available vision or other senses, primary aids such as white cane, or secondary aids such as other indoor navigation tools. However, the presence of landmarks is considered as a parameter in route planning in only four studies [21,26,84,85].

When dealing with obstacles, avoiding areas with obstacles completely is the most frequent strategy. The avoided objects can serve the purpose of an obstacle and a landmark both. When the route planning strategy avoids obstacles completely, rather than adjusting the path to safely detour the obstacle, there may be less information for the user to support orientation and developing mental maps.

While this review reveals several factors adopted for determining optimum routes in BVI indoor navigation tools, two key limitations related to such factors can be identified: (1) adopting a single factor for route optimization while many factors simultaneously contribute to optimize routes, and (2) when integrating several factors, individual factors are prioritized in an *ad-hoc* manner. More than one factor to determine optimum routes are adopted by 33 articles; however, 18 of them consider distance and obstacles (pre-avoiding, weight-adjusted or real-time evidence) only. Therefore, it is evident that multi-criteria route planning is still not prominently addressed in the BVI indoor navigation tools. Weights for the different factors are identified based on a user survey in only a single study [26].

Key findings: RQ2 – addressing user variations in route planning

It can be assumed that all the articles which concern not only the route length, including the articles which avoid obstacles in initial route identification, acknowledge the differences between sighted and BVI navigation. However, even when the BVI users are provided with navigation routes that are different than those for sighted persons, the majority of the articles treat the BVI community as a general group, whereas in reality navigation requirements of different BVI persons vary based on different factors. When the user is generalized to a single group, the individual user context is lost and the route would affect the navigation experience. For example, the user is assumed as using a white cane by Dong and Ganz [28] but not all BVI persons are white cane users. Only 15 articles (11%) have attempted to provide routes considering different user capabilities.

Key findings: RQ3 – procedures and algorithms used for route planning

Standard route planning algorithms such as Dijkstra's shortest path algorithm and A* algorithm are frequently adopted for indoor route planning of BVI navigation tools. The use of heuristics such as turns or the Manhattan distance has reduced the search space when A* algorithm is used. For dynamic route planning, A* and D* algorithms are commonly selected.

Unique needs of the BVI community such as wall-following support cannot be integrated into the routes without adjusting the search space of the algorithms to identifying routes close to walls. The most common variation in the search space adopted is excluding the fixed obstacles in the environment. Finding routes close to walls is included in the route planning procedure in five studies. Novel route planning algorithms which incorporate several unique BVI requirements are introduced by Swobodzinski and Raubal [21] and Ivanov [26]. Ivanov [84] and Ivanov [85] have extended the route planning method in [26] and incorporate user and building information into it.

The route calculation by Swobodzinski and Raubal [21] is developed based on the idea that if the route can consider the principles of white cane use, it would be better for a blind person in terms of safety and orientation than a route, which considers only distance minimization. An algorithm, which finds routes supporting wall-trailing, avoiding obstacles and having landmarks and cues along the route is developed in [21]. The algorithm by Ivanov [26] also finds routes supporting wall-trailing and avoiding obstacles and then refine it with more parameters. The main drawbacks of the approach in [21] are, not considering different user needs and difficulty to integrate other BVI needs. How multi-floor areas can be traversed with the procedure needs further exploration. The multiple factors adopted in [26] are prioritized based on a user study however the weights are assumed to be universal. Both Swobodzinski and Raubal [21] and Ivanov [26] have assumed that all BVI persons prefer routes close to walls, which limits the usability of the approaches. Real-time obstacle avoidance is addressed in several studies, however, the majority of such studies do not address other BVI navigation needs in route planning.

Though veering is a common problem for BVI persons, recovery from veering is considered in a limited number of studies (16 out of 131). Re-routeing in real-time to cater to dynamic obstacles is addressed in 14 articles. However, when addressed, the re-routeing aspect is covered to a greater extend with implementations in a majority of these articles. Computer vision-based approaches are commonly adopted for real-time obstacle de-touring. The studies on real-time obstacle de-touring without integrating to route planning is not included in the scope of this review, and the results have to be interpreted accordingly.

Through there are challenges and difficulties in navigating inside buildings, BVI persons already have navigation skills supported by O & M techniques. Overall, the route planning methods paid less consideration for the already developed skills of the user. Letting the user find the route using their conventional strategies while providing additional information and guidance is one significant approach in this regard as explored by Oh et al. [95].

Key findings: RQ4 – building data modelling for route planning

Both grid-based and network-based modelled are frequently adopted for modelling the indoor environments in the selected articles. In the absence of pre-defined routes like a road network in the outdoor, the indoor route networks have to be first defined to derive optimum routes using different algorithms. The way the indoor spaces are modelled affects the route planning procedures of the BVI navigation tools. Obstacles are commonly modelled so that they can be avoided in route planning. Though landmarks provide vital support to navigation, modelling landmarks of different forms is very limited. Simple network models reduce the building information to graphs connecting doors or rooms. The visibility graph approach is used in a few articles, which supports obstacle avoidance.

In order to model routes close to walls, a simple room to room network approach is changed in few studies [21,26,32,33,47,48]. Walls and obstacles are included in the building models as key features by Swobodzinski and Raubal [21], and Ivanov [26] to generate buffers and thereby routes close to walls and away from obstacles.

Semantic information of the buildings such as landmarks, staircases, and lighting levels are also important for BVI navigation. Though semantic building information useful for BVI route planning is identified in few studies [26,42,84,85,131], how such semantic information is integrated to route planning is not presented in detail. Light levels can strongly affect the navigation of BVI persons with limited contrast sensitivity, however, such environment information is not considered in the majority of the tools. Incorporating dynamic changes in the environment is considered in a sub set of studies, which have addressed the real-time obstacle avoidance.

Standard building models provide more support relate to modelling semantic, topological and geometry information together so that routes rich with context information can be derived. However, standard building data models are used rarely in the selected articles.

Many applications use small, simple or structured floor areas as the sample areas when designing the route planning methods; however, in the real world, the assistance of navigation tools is mostly required in complex environments. Simple sample areas would be sufficient for verifying the micro-navigation support of the tools but would not cover challenges in macro-navigation in large public spaces (e.g., government offices, hospital) such as facing human traffic, floor transition, need for taking many turns, changing lighting conditions and various types of obstacles. Few applications [46,75,86] have considered complex environments such as multi-floor navigation and large crowded areas with complex floor layouts.

Suggestions for the assistive technology domain

Based on the findings of this study, the main directions for enhancement of the route planning modules of BVI indoor navigation tools can be identified as follows:

RQ1 and RQ2: factors affecting optimum routes and integrating user differences

Though many parameters are considered in deriving optimum routes for the BVI persons as shown in Table 2, how to trade-off between them needs to be further studied, as there is a notable gap in the literature regarding this aspect. The identified factors can be further verified against current BVI needs. It is important to consider the user perspective as a major concern. Assistive technology developers need to work closely with the end-users and the experts in the O & M domain to identify user requirements related to indoor routes in modern complex buildings. End-user surveys on BVI navigation done with the BVI communities, input from O & M instructors, and O & M guidelines can provide useful insight for identifying the actual requirements for BVI communities expected in macro-navigation support as a whole.

BVI people are a community with considerable variability in their range of sight abilities [138]. Though mobility is greatly affected by visual impairments, whether it has resulted from changes in visual acuity, visual fields, depth perception or contrast sensitivity (Bibby et al., Lord and Dayhew, Marron and Bailey as cited by [139]), the way these conditions affect the navigation abilities are not the same as the space perception is impacted differently. Visual field and contrast sensitivity are more importantly contributing to orientation mobility skills than visual acuity [140]. Out of the total population of visually impaired people in 2010, only 14% are blind and 86% have a low vision [141]. More than 50% of the persons with vision impairment is reported to have one or more other impairments also (Kirchner, as cited by [142]), which can further affect their navigation and mobility abilities. Therefore, user variations can be occurred due to various reasons such as (a) primary assistive device use: e.g., using a white cane, guide dog, both of them or none, (b) visual impairment: e.g., totally blind, partially blind or see in good lighting conditions, (c) other limitations: e.g., using a wheel-chair, and (d) preferences: e.g., some may prefer to use/avoid staircases or escalators; even though the wall–trailing may be preferred by many BVI individuals, some may not. The navigation of a BVI person is not mechanical, robot-like travelling; there are already established navigation strategies used by the BVI persons with various levels of limitations. It is better to complement the already familiar strategies of the BVI persons by the navigation tools, instead of substituting them, so that the users feel comfortable and empowered. Therefore, it is recommended to revisit the design of the BVI indoor navigation tools to integrate customization of route planning parameters based on user preferences as a vital component. An approach where several generalized user groups and set of preferences related to route determination is provided by the tools is suggested so that users can integrate their preferences when using the tool. The needs of the BVI users including the route planning aspect of indoor navigation tools are explored and valuable inputs are provided by several studies [143–147], which can provide inputs for future designs.

RQ3: procedures and algorithms

Some unique requirements generally applicable in BVI navigation based on O & M guidelines are: (a) supporting wall-trailing, (b) avoiding crossing large spaces, (c) support detouring fixed and dynamic obstacles and recovery from veering, (d) supporting paths with less turns, and (e) informing landmarks and obstacles in or close to the path. In order to derive and update routes satisfying such general requirements, the search space of the algorithms need to be adjusted (e.g., first selecting paths close to walls and then optimize those path). Therefore, the modelling of the indoor navigation routes becomes an important concern for BVI route planning. The route planning procedure can be performed basically in two steps: (1) identifying navigable routes satisfying BVI persons (finding the search space), and (2) finding the optimized routes considering multiple factors and user preferences. Failure to identify the suitable search space limits the optimality of the routes. Therefore, investigating procedures to model the indoor routes suitable for BVI users is recommended.

Key limitations of the procedures adopted for route planning are (a) difficulty in satisfying multiple BVI navigation needs simultaneously by the selected approaches, and (b) failure to adopt for user variations. Therefore it is recommended to treat the indoor route planning for BVI navigation as a user-centric, multi-objective optimization problem so that better route planning algorithms can be derived.

When optimizing the routes, standard routeing algorithms such as Dijkstra's algorithm can be executed on the navigable routes first derived considering BVI needs while A* algorithm or D* algorithm would be better suited for dynamic route planning. Comprehensive route planning procedures, however, would satisfy several criteria such as: (1) integrate above two route planning steps together, (2) cater for multiple parameter optimization,

(3) support user variations, (4) manage real-time obstacle detouring, (5) support recovery from veering, and (6) suitable to deploy in complex building environments. While different articles present useful approaches to address some of these requirements, more research is required to integrate many of these requirements in a single route planning module. Some notable approaches which can be usefully related to such integration are: incorporating unique BVI navigation requirements in route planning methods [21,25,84,109], 3D real-time obstacle avoidance [24,70], modelling building features in a way useful for BVI navigation [20,21,85,106], supporting navigation in a complex environment [46], use of landmarks for route verification [52,57,68,95], addressing user variations [25,85,109], use of interactive tactile maps at kiosks [127,128].

RQ4: building data modelling for route planning

Indoor routes are derived based on indoor structures and their attributes, and to model the routes suitable for BVI users, the modelling of the building data has to be done from the BVI navigator perspective. It is recommended to look at the building data from the BVI navigation perspective so that the relevant building data can be identified and converted to suitable representations. The geometric, topological and semantic aspects of the data need to be sufficient enough to generate optimum routes. For example, if the wall-trailing is to be supported by the route planning methods, information about walls, which is not considered important for sighted users, needs to be an integral part of the building model; what constitutes obstacles or indoor landmarks needs further exploration to provide semantic information. Another concern would be the dynamic nature of the building environment, which would result in a frequent need to update the data model. A model, which represents fixed structures initially with a capability to integrate modifications such as placement of furniture dynamically would lead to generating realistic routes.

Exploring the possibility of adopting standard building models to represent BVI users' needs is recommended, as it will support make use of the existing spatial data of the indoor environments, utilizing already available functionalities provided by the standard data models, and re-producing the applications in other environments easily. Analyzing the suitability of adopting IndoorGML, CityGML-LO4 (https://www.ogc.org/standards/citygml), OSM with indoor tagging (https://wiki.openstreetmap.org/wiki/Simple_Indoor_Tagging), BIM and CAD-based approaches in BVI indoor navigation is suggested. Integration of both these aspects, i.e., considering specific BVI needs and adopting standard building models together, will result in models supporting BVI route planning while offering all capabilities of standard building models. For examples of using standard building models and extending them regarding BVI indoor navigation, which can be considered for future enhancements, see [24,38,48,90,100,106,112].

BVI navigation prototypes, after verifying the concept with a simple sample scenario, are recommended to deploy in large and complex building settings which reflect the real world to understand the design limitations and challenges. Attempts to transferring technology to practice will reveal the challenges in modelling complex buildings for BVI route planning and navigation.

Not only the route planning, other aspects of BVI indoor navigation tools, such as interpreting the indoor positions with respect to the surrounding and providing orientation support, also depend strongly on building information; what features and the level of details of the building features to model, and how they are modelled affect such functionalities. Therefore, modelling building data from the BVI navigation perspective and adopting standard building models would support other aspects of the navigation tools as well.

Limitations of the study

The objective of the review is to identify what route planning strategies were adopted by the BVI indoor navigation systems. Therefore the articles are selected irrespective of the state of the route planning methods they presents; they may present a concept, a design, a prototype and/or real-world implementations. Therefore, the route planning strategies identified can be at the conceptual or design stage in some cases while others present systems with implementation as a prototype or real-world deployments. The impact of excluding the studies presenting only the concepts or designs is not investigated in this study. Whether the strategies adopted by the articles are evaluated by BVI users or not is also not considered. Therefore, the effectiveness of the approaches in a real-world scenario cannot be inferred from the results. Articles that present the continuation of the same project are not excluded and therefore the number of articles selected for the review does not uniquely represent a number of route planning strategies reviewed. Information related to the time complexity and the performance of the route planning algorithms were not included in this review. Though a comprehensive search strategy was employed in 13 databases, no articles are identified through the reference lists of the selected articles or from the grey literature.

Conclusion

In this systematic review, 131 primary studies have become eligible based on a comprehensive search on multiple data sources covering 1982–July 2018. Among them, sufficient details of factors affecting route planning are given by 87 studies. More than half of the studies optimize only the distance, thereby failed to incorporate any other BVI navigation requirements. Obstacles, turns, landmarks, wall-trailing support, and cognitive load to travel are other key factors adopted to accommodate BVI needs. User variations are addressed in a limited number of studies by (1) allowing the individual users to provide preferences, and/or (2) defining rules based on user groups. Information about the algorithms or the planning methods is presented in 79 studies. The use of Dijkstra's shortest path algorithm with distance minimization is prominent (42%).

Less consideration for the unique needs of the BVI community and less concern on addressing variations in user needs are the main limitations in the route planning methods identified. The key recommendation to improve the route planning components of BVI indoor navigation tools, based on the results of the study, is to integrate user perspective strongly in all aspects of route planning. The findings of this study will help to frame future research and development activities on designing better assistive tools for BVI indoor navigation, particularly route planning modules, aligning with various user requirements and O & M guidelines.

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Data availability statement

The research protocol and the list of eligible records of this study are openly available in figshare repository at https://doi.org/10.6084/m9.figshare.12821759 and https://doi.org/10.6084/m9.figshare.12821786.v3 respectively.

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